

Frequency-domain training data for a learned plasma fluid closure with an extension to the $N = 2$ even-moment closure and Landau damping

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Abstract

We close the warm-beam moment hierarchy of a cold-bulk / warm-beam bump-on-tail system with a small MLP that replaces the Hammett–Perkins / Hunana Padé rational closure of $Z'(\xi_b)$. The architecture is standard; the contribution is the training-data design. The default approach — run many linearized kinetic simulations across a $(u_b/v_b, \varepsilon, k)$ grid and harvest (ratios, α) pairs from trajectories — gives a closure that is already a respectable improvement over Padé: γ_{eff} within $\sim 10\%$ across most of the parameter grid, degrading to $\sim 50\%$ at the marginal corner. Sampling ξ_b directly in a $\mathbb{C}$ rectangle — the equivalent frequency-domain operation, evaluated in closed form via the Faddeeva function — gives the same NN 10^{-4} – 10^{-3} relative error nearly everywhere, two decades tighter than the sim-trained closure and three to four decades tighter than Padé. The frequency-domain dataset is also $\sim 100\times$ cheaper to generate (one Faddeeva eval per sample, no kinetic time-stepping, no IC choices) and is problem-agnostic in $(u_b/v_b, \varepsilon, k)$. The geometric reason the win exists: linear eigenmodes from a kinetic simulation trace a narrow region in the ξ_b -plane, leaving the rest of the relevant rectangle uncovered.

1 Setup

A cold background plasma (density n_0) coupled to a warm Maxwellian beam (density $n_b = \varepsilon n_0$, drift u_b , thermal speed v_b). In beam-frame coordinates ($s = (v_x - u_b)/v_b$) the linearized two-fluid system has state $y = (u, E, U_0, U_1, U_2)$ with

$$\partial_t u = -E, \tag{1}$$

$$\partial_t E = u - \varepsilon(u_b U_0 + U_1), \tag{2}$$

$$\partial_t U_n = -iku_b U_n - ikU_{n+1} + nM_{n-1}E, \quad n < 2, \tag{3}$$

where $U_n = (1/n_b) \int s^n f_{b1} ds$ and $M_k = \langle s^k \rangle_{f_{b0}}$ ($M_0 = 1$, $M_1 = 0$, $M_2 = 1/2$, ...). The closure problem is to specify U_3 at each step.

The kinetic linear response of the beam to a potential $\hat{\Phi}$ is

$$U_0/\hat{\Phi} = Z'(\xi_b), \quad \xi_b = (\omega - ku_b)/(kv_b), \tag{4}$$

with $Z(\xi) = i\sqrt{\pi} w(\xi)$ the plasma dispersion function and w the Faddeeva function. Iterating the recurrence $\xi_b U_n - U_{n+1} = nM_{n-1}\hat{\Phi}$ from (4) gives the exact 3-moment closure ratio

$$\alpha(\xi_b) \equiv U_3/U_0 = \xi_b^3 - \xi_b Z'(\xi_b). \tag{5}$$

The kinetic dispersion at scales much longer than the Debye length is

$$D_{\text{kin}}(\omega, k) = 1 - 1/\omega^2 - (\varepsilon/k^2)Z'(\xi_b) = 0. \tag{6}$$

Padé closures. A linear-in-moments closure $U_3 = \sum_{i<3} a_i U_i$ with constants a_i combined with the moment recurrence yields a degree-(1, 3) rational approximant of $Z'(\xi_b)$. Choosing (a_i) to match the Taylor series of $-\sqrt{\pi}Z'$ at $\xi_b = 0$ to order ξ_b^N is the Padé closure R_N in the Hammett–Perkins / Hunana hierarchy.

2 Padé baseline

Figure 1 compares the closed fluid response of the Padé hierarchy $N \in \{3, 4, 5, 6\}$ to the kinetic $-\sqrt{\pi}Z'(\xi_b)$ along real ξ_b . Each Padé closure matches the kinetic curve to $\sim 10^{-2N+3}$ at $\xi_b = 0$ (Taylor matching) and degrades rapidly to residual of order unity for $|\xi_b| \gtrsim 1$. The imaginary part (Landau resonance) is captured at only $\sim 30\%$ of its kinetic amplitude even at $N = 6$. This is the fundamental Padé ceiling: a finite-order rational approximates an entire function only locally.

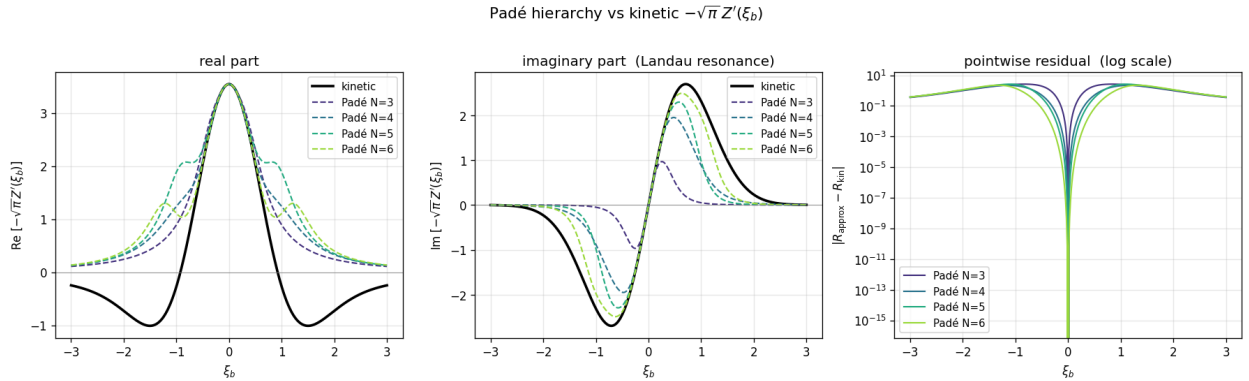


Figure 1: Padé hierarchy vs kinetic $-\sqrt{\pi}Z'(\xi_b)$ along real ξ_b . Real part (left), imaginary part / Landau resonance (middle), pointwise residual on log scale (right). Padé residuals plateau at $\sim 10^0$ for $|\xi_b| > 1$.

Figure 2 plots the relative error in the BoT $\gamma_{\max}$ at the canonical operating point ($u_b/v_b = 5, \varepsilon = 0.05$) vs Padé order. The hierarchy oscillates around the kinetic answer, crossing zero between $N = 7$ and $N = 8$.

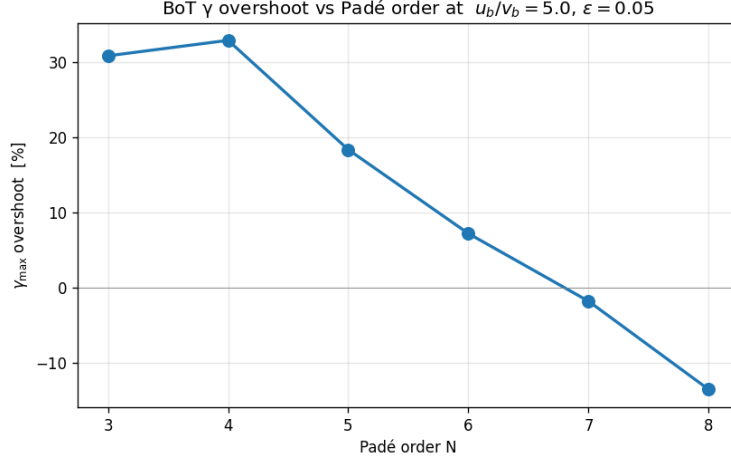


Figure 2: $\gamma_{\max}$ overshoot at $(u_b/v_b, \varepsilon) = (5, 0.05)$ vs Padé order. Non-monotone — crosses zero near $N = 7$.

3 A closed-form identity for the per-mode closure

Before turning to the NN we record one closed-form identity. It fixes what *any* closure must reduce to on a single linear eigenmode, and will let us state precisely what the NN does and does not learn (Section 9).

Applying the moment recurrence $U_{n+1} = \xi_b U_n - n M_{n-1} \hat{\Phi}$ for $n = 0, 1, 2$ with $M_0 = 1$, $M_1 = 0$:

$$\begin{aligned} U_1 &= \xi_b U_0, \\ U_2 &= \xi_b U_1 - M_0 \hat{\Phi} = \xi_b^2 U_0 - \hat{\Phi} \quad (\text{since } M_0 = 1), \\ U_3 &= \xi_b U_2 - 2M_1 \hat{\Phi} = \xi_b U_2 \quad (\text{since } M_1 = 0). \end{aligned}$$

The third line eliminates the inhomogeneous coupling because the first velocity moment of a Maxwellian vanishes in its own frame. Dividing by U_0 and using $\xi_b = U_1/U_0$:

$$\boxed{\alpha = \frac{U_3}{U_0} = \frac{U_1}{U_0} \cdot \frac{U_2}{U_0}} \quad (7)$$

i.e. on any linear eigenmode the closure ratio is the complex product of the two moment ratios. Direct check: $r_1 r_2 = \xi_b (\xi_b^2 - 1/Z'(\xi_b)) = \xi_b^3 - \xi_b/Z'(\xi_b) = \alpha(\xi_b)$. All the plasma-dispersion machinery is absorbed into U_2 ; the closure step itself is multiplication.

Used as a closure, this fails. On a superposition $U_n(t) = \sum_i A_i e^{-i\omega_i t} U_n^{(i)}$ the identity breaks: the product of sums in $r_1 \cdot r_2$ generates cross-terms ($i \neq j$) that the per-mode formula $\sum_i A_i \xi_{b,i} U_2^{(i)}$ does not. So hardcoding $U_3 = U_1 U_2 / U_0$ in a fluid solver is not a usable closure. At the canonical $(5, 0.05, k = 0.24)$ point, with a Langmuir-projected IC (single eigenmode), bilinear matches the kinetic γ_{eff} to six decimals; with a generic δE IC (all five fluid eigenmodes) it gives +28.4% overshoot — essentially Padé-level (+30.8%).

The same picture holds across the full sweep (Figure 3, middle panel): bilinear is a strict improvement on Padé in every cell, but the improvement is small (8–14% relative in the regimes where Padé is acceptable; larger only at the marginal corner where both are deep into failure). In

practical terms it is not a useful alternative to Padé. Padé and the Hunana hierarchy chose linear-in-moments closures precisely so that they commute with superposition; (7) breaks that property and inherits the cost.

Why we record it anyway. The identity (7) fixes what the per-mode closure equals, in closed form. The NN closure ($f_\theta : (r_1, r_2) \mapsto \alpha$) is structurally free to be anything; the identity gives a baseline against which f_θ can be compared on and off the eigenmode manifold. Section 9 uses it to show that the NN reproduces the bilinear product on the manifold but a distinctly non-multiplicative function off it, and that the off-manifold behaviour is what makes the closure usable.

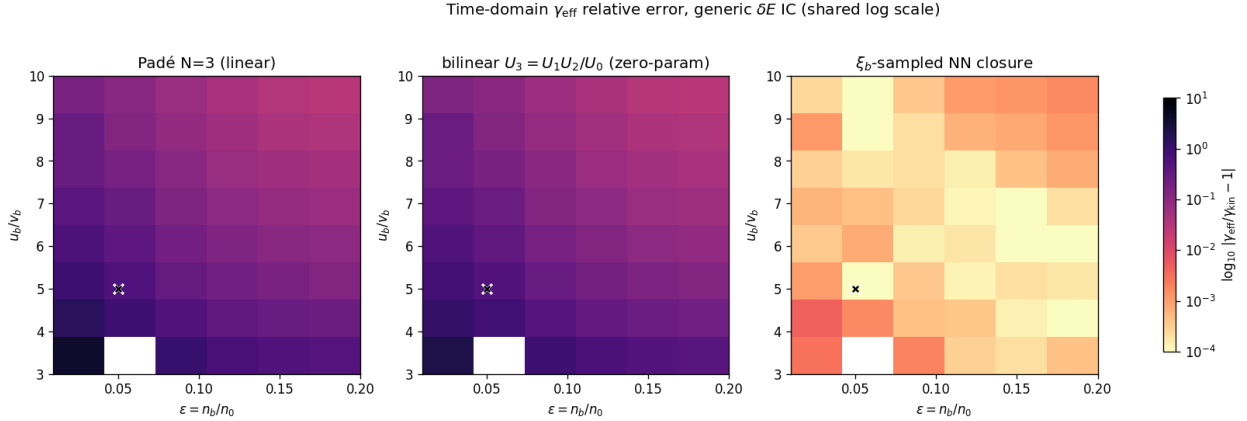


Figure 3: Time-domain γ_{eff} relative error across the $(u_b/v_b, \varepsilon)$ grid from a generic δE IC, on a shared log scale. *Left*: Padé $N = 3$. *Middle*: bilinear identity $U_3 = U_1 U_2 / U_0$ used as a closure (Section 3). *Right*: ξ_b -sampled NN closure (Section 8). The bilinear identity is a strict but small improvement on Padé in every cell — both are in the 10^{-1} – 10^0 regime. The NN sits two to three decades tighter.

4 NN closure

We replace the Padé linear closure with a small MLP $f_\theta : (U_1/U_0, U_2/U_0) \rightarrow \alpha$ that outputs the closure ratio directly:

$$U_3 = f_\theta(U_1/U_0, U_2/U_0) \cdot U_0. \quad (8)$$

This is a nonlinear-in-moments closure with no rational-order ceiling: NN expressivity bounds the approximation error, not the polynomial degree. The architecture is fixed throughout: 4-layer MLP, (64, 64, 64, 64) hidden, tanh activations, $\sim 8.7\text{k}$ parameters. Input: 4 reals (re/im of two complex moment ratios). Output: 2 reals (re/im of α). Adam, cosine LR $3 \times 10^{-3} \rightarrow 10^{-5}$, 8000 steps, batch 512. What changes between the experiments below is the *training data*.

5 First attempt: simulation-trained closure

The default training-data choice is the same one used everywhere else in data-driven closures: integrate the kinetic system, harvest moments (U_0, U_1, U_2, U_3) along trajectories, regress U_3/U_0 on $(U_1/U_0, U_2/U_0)$. We build a broad sweep: 24 operating points $(u_b/v_b \in \{3, 4, 5, 6, 8, 10\}) \times (\varepsilon \in \{0.02, 0.05, 0.10, 0.20\})$, five k values per operating point, 12 random initial conditions each.

Total: $\sim 10^5$ training samples after filtering transient outliers. The sweep is intentionally broad to maximise coverage. This is a substantial amount of effort: $24 \times 5 = 120$ kinetic integrations, with mode-projection bookkeeping at each timestep.

Linear dispersion. Figure 4 compares the sim-trained NN to the kinetic $\gamma(k)$ at four representative $(u_b/v_b, \varepsilon)$ points. The closure tracks the kinetic dispersion curve through the unstable branch at three of the four; peak- γ overshoot is below 1% near the centre of the training set and grows to +12% at the marginal corner $(3, 0.05)$. The spikes in damped- k regions are artefacts of fitting γ from a generic-IC trajectory where no unstable mode exists — the late-time signal is dominated by whichever fluid eigenmode is least damped, and the sim-trained closure produces inaccurate spurious modes there.

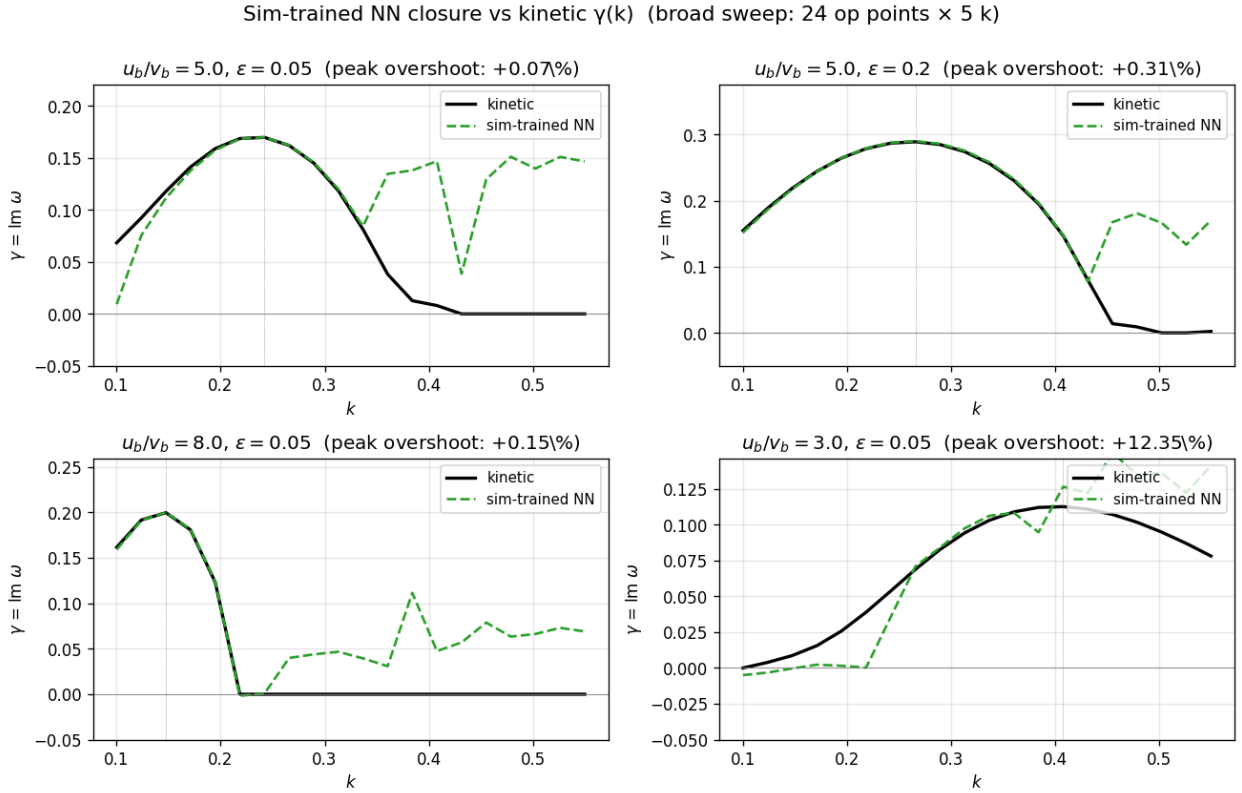


Figure 4: $\gamma(k)$ for the simulation-trained NN closure (green dashed) vs kinetic (black) at four representative operating points. The sweep spans the training distribution; dispersion match is $\lesssim 1\%$.

Time-domain at canonical. Figure 5 integrates the sim-trained closure at the canonical operating point under both a Langmuir-projected and a generic- δE IC. The closure tracks the kinetic envelope to graphical precision in both cases; fitted $\gamma_{\text{eff}} = +0.1701$ vs $\gamma_{\text{kin}} = +0.1700$.

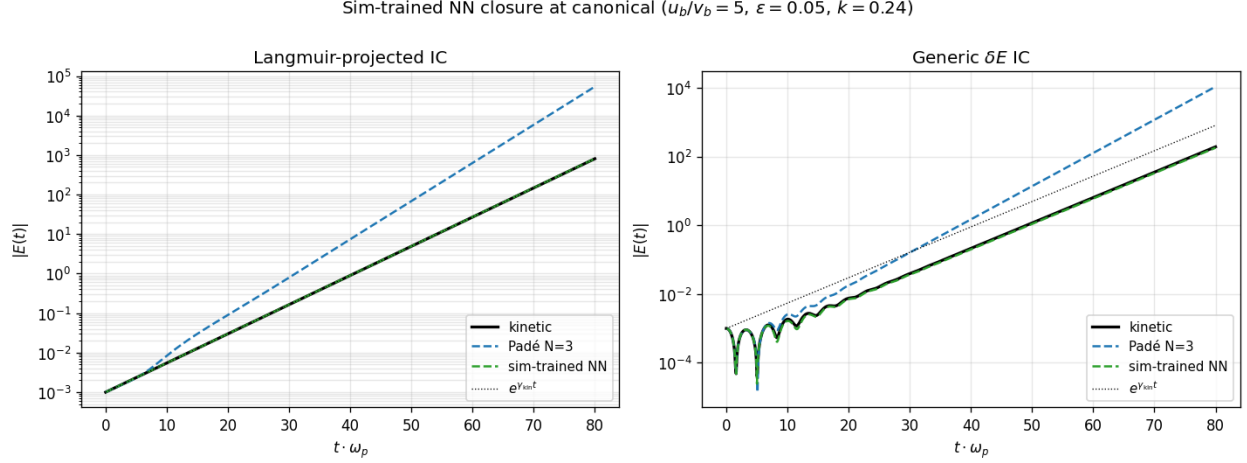


Figure 5: Sim-trained NN closure at canonical ($u_b/v_b = 5$, $\varepsilon = 0.05$, $k = 0.24$). Tracks the kinetic envelope under both ICs — this operating point is well within the training distribution.

2D parameter scan. Figure 6 shows the time-domain γ_{eff} relative error from a generic δE IC across the $(u_b/v_b, \varepsilon)$ grid. The colour scale is $\log_{10} |\gamma_{\text{eff}}/\gamma_{\text{kin}} - 1|$ from 10^{-4} to 10^1 , shared with Figures 8 and 10. Padé sits at 10^{-1} – 10^0 . The sim-trained NN is roughly one decade better than Padé in the well-trained interior of the grid and stays at 10^{-2} – 10^{-1} over most cells, with the marginal corner reaching $\sim 10^{-0.3}$. This is genuinely a serviceable closure — within $\sim 10\%$ on γ_{eff} across most of the operating regime, which is what one would hope to get from a data-driven approach trained on 10^5 simulation samples.

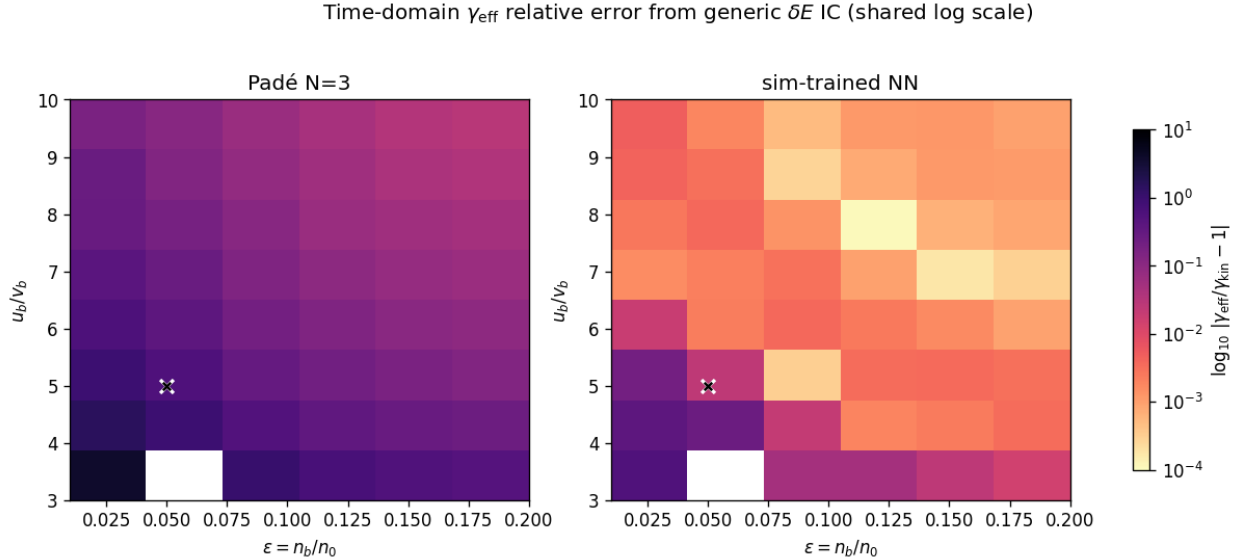


Figure 6: Time-domain γ_{eff} relative error for Padé (left) and the simulation-trained NN closure (right) on a shared log scale. The sim-trained closure is a decade better than Padé in the well-trained interior, with the largest residuals at the low- ε low- u_b corner.

The natural question is whether one can do better without simply running more simulations.

6 Diagnosis: 1D coverage in ξ_b

The closure problem is fundamentally one-parameter: on a linear eigenmode, α is a function of ξ_b alone (5), and $(U_1/U_0, U_2/U_0)$ are too. So we can plot any training dataset as a distribution in the ξ_b -plane.

Figure 7 does this. The orange rectangle is the ranges of ξ_b relevant for unstable BoT modes ($\text{Re } \xi_b \in [-3, 3], \text{Im } \xi_b \in [-1, 1.5]$). The green scatter is the ξ_b value on the most-unstable kinetic mode as $(u_b/v_b, \varepsilon, k)$ is varied across a fine grid covering and exceeding the simulation training set. The scatter occupies a narrow sub-region of the rectangle: it is bounded to $\text{Im } \xi_b > 0$ (unstable modes have positive Landau-resonance imaginary part) and is heavily concentrated near a 1-parameter curve in the upper-left quadrant.

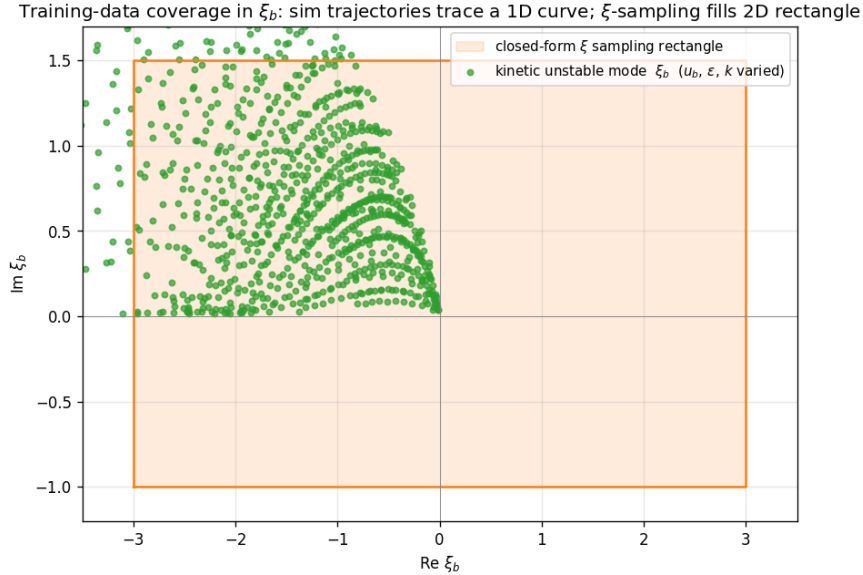


Figure 7: ξ_b on the most-unstable kinetic mode as $(u_b/v_b, \varepsilon, k)$ is varied (green); closed-form ξ_b -sampling rectangle (orange). The orange rectangle covers all relevant ξ_b including damped/spurious modes; the green region covers only the unstable branch.

Why this matters. The 5-d closed fluid system has *five* eigenmodes, only one of which is the Langmuir-like unstable mode. A generic- δE IC excites a superposition of all five. Their projected ξ_b values are spread across the rectangle — many of them outside the green region. When the NN sees moment ratios from those modes it has no training data nearby and outputs an extrapolated α . If the extrapolated α happens to overstabilise the spurious mode the trajectory still settles onto Langmuir-like growth (this is why canonical single-trace tests look fine); if it doesn't — as in the marginal corner — the spurious modes contaminate γ_{eff} .

This is structural. No amount of additional simulation training data fills the rectangle: kinetic trajectories live on the eigenmode manifold, and the eigenmode manifold's projection in ξ_b -space *is* the green region. Adding more $(u_b/v_b, \varepsilon, k)$ adds more samples on the same set.

7 Fix: ξ_b -sampling

The kinetic linear theory provides an analytical handle. On a linear eigenmode the forward map $\xi_b \mapsto (U_1/U_0, U_2/U_0)$ is closed-form: starting from $U_0/\hat{\Phi} = Z'(\xi_b)$ and iterating the moment recurrence,

$$U_0 = Z'(\xi_b), \quad U_1 = \xi_b Z'(\xi_b), \quad U_2 = \xi_b^2 Z'(\xi_b) - 1, \quad (9)$$

$$U_1/U_0 = \xi_b, \quad U_2/U_0 = \xi_b^2 - 1/Z'(\xi_b). \quad (10)$$

Combined with (5), each tuple $((\text{Re}, \text{Im } U_1/U_0, \text{Re}, \text{Im } U_2/U_0), (\text{Re}, \text{Im } \alpha))$ is one Faddeeva evaluation. *Same equations as the simulation training; same single underlying kinetic linear theory.* The only difference is that we sample ξ_b directly (frequency-domain) rather than reconstructing it from time-domain trajectories.

We replace the simulation dataset with 16,384 ξ_b values uniformly distributed in $\text{Re } \xi_b \in [-3, 3]$, $\text{Im } \xi_b \in [-1, 1.5]$ — the orange rectangle in Figure 7. The training and architecture are otherwise identical.

Data-generation cost. The simulation dataset of §4 required 120 kinetic integrations across the operating sweep ($\sim 10^5$ filtered samples; minutes of wall time on a single core including the mode-projection bookkeeping). The closed-form dataset is 16,384 `scipy.special.wofz` calls — a fraction of a second of wall time, no operating-point sweep, no ICs, no transient outliers to filter. The ratio is roughly two orders of magnitude in compute and a much larger ratio in design effort.

α accuracy at test points. Table 1 compares α prediction error at representative ξ_b values for the sim-trained (single op pt), sim-trained (broad, 24 op pts) and closed-form-trained closures. The first column is well inside all training distributions and all three closures work. The remaining rows are off the unstable-mode curve — moment ratios from damped or spurious modes. Sim training fails by 50–800%; closed-form training is uniformly sub-1%.

ξ_b	sim (1 op pt)	sim (broad, 24 op pts)	closed-form
$-1.17 + 0.71i$ (canonical eigenmode)	+2.3%	+0.8%	+0.05%
$-0.83 + 0.05i$	+148%	+62%	+0.5%
$0.50 + 0.00i$	+843%	+247%	+0.6%
$-1.50 - 0.20i$	+50%	+27%	+0.2%
$1.20 + 0.10i$	+209%	+118%	+0.6%
$-2.00 + 0.00i$	+13%	+7%	+0.5%

Table 1: $|\alpha_{\text{NN}} - \alpha_{\text{exact}}|/|\alpha_{\text{exact}}|$ at representative ξ_b . Sim sampling fits well only at the unstable Langmuir mode; closed-form is accurate everywhere.

8 Results with ξ_b -sampling

Linear dispersion sweep. Figure 8 compares the γ_{max} relative error $|\gamma_{\text{fluid}} - \gamma_{\text{kin}}|/\gamma_{\text{kin}}$ across the $(u_b/v_b, \varepsilon)$ grid on the same shared log scale as the previous section. Padé $N = 3$ sits at 10^{-1} – 10^0 (worst at marginal, matching Figure 2). The ξ_b -sampled NN closure is $\leq 10^{-3}$ across nearly the entire grid — two to four decades tighter than the sim-trained closure (Figure 6) and three to four decades tighter than Padé.

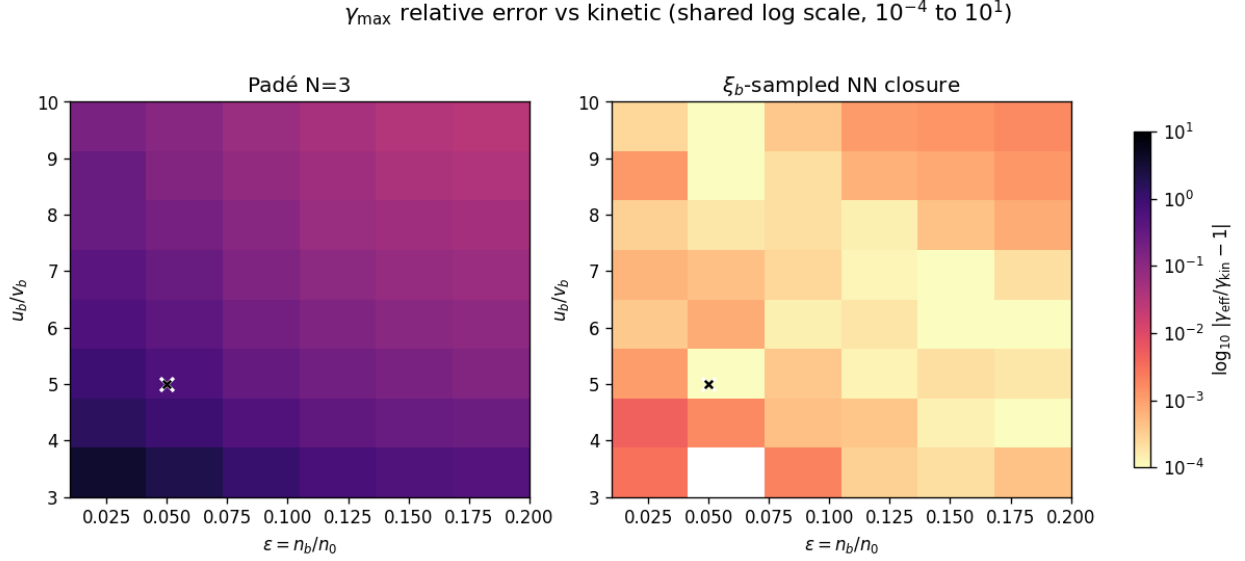


Figure 8: $\gamma_{\max}$ relative error vs kinetic across the $(u_b/v_b, \varepsilon)$ grid. Padé $N=3$ (left) vs ξ_b -sampled NN closure (right) on the shared log scale. White $\times$ marks canonical (5, 0.05).

Table 2 reports four representative points.

$(u_b/v_b, \varepsilon)$	γ_{kin}	Padé $N=3$	ξ_b -sampled NN
canonical (5, 0.05)	0.170	+30.8%	−0.02%
hot beam (5, 0.20)	0.289	+13.9%	+0.02%
strong drift (8, 0.05)	0.200	+11.2%	+0.01%
marginal (3, 0.05)	0.113	+107.9%	+0.20%

Table 2: $\gamma_{\max}$ overshoot at representative operating points. $\sim 10^3$ – $10^4\times$ improvement in relative error vs Padé $N=3$.

Time-domain canonical stress test. Figure 9 integrates all three systems for $t \in [0, 80/\omega_p]$ at the canonical operating point, with Langmuir-projected and generic- δE ICs. Fitted γ_{eff} : $\gamma_{\text{kin}} = +0.1700$, $\gamma_{\text{pade}} = +0.2223$, $\gamma_{\text{NN}} = +0.1700$.

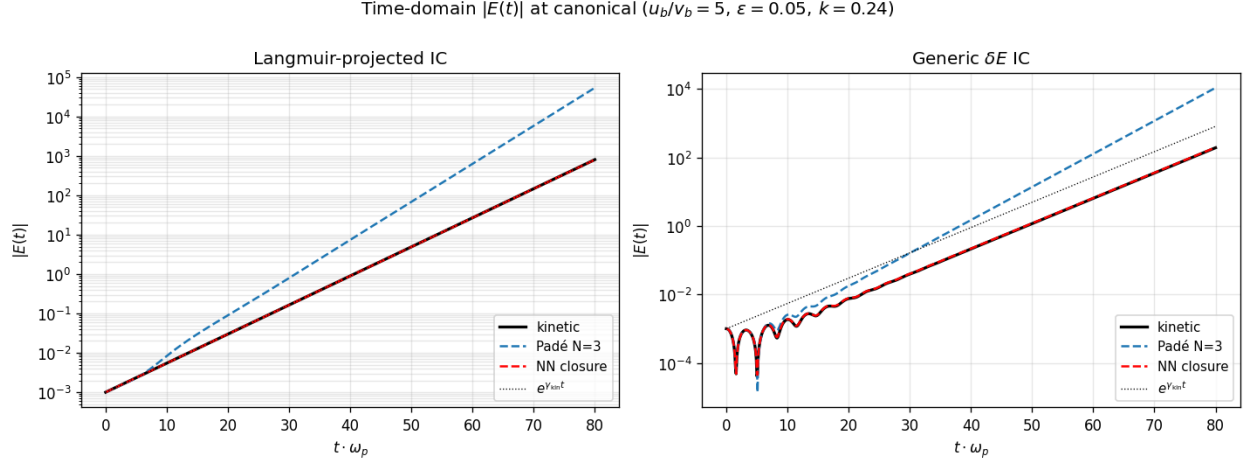


Figure 9: Time-domain $|E(t)|$ at canonical ($5, 0.05, k = 0.24$) under Langmuir-projected and generic δE initial conditions. ξ_b -sampled NN closure (red dashed) tracks kinetic (black) in both cases.

Time-domain across parameter space. Figure 10 repeats the experiment for each $(u_b/v_b, \varepsilon)$ cell. The ξ_b -sampled closure is $\lesssim 0.1\%$ over most of the grid; the marginal corner shows mild under-prediction (NN training rectangle's lower-left isn't quite reached in ξ_b). No blow-up anywhere. The contrast with Figure 6 is the headline: the same NN architecture, trained on the same physics but sampled in frequency rather than time, fixes the marginal-corner failure.

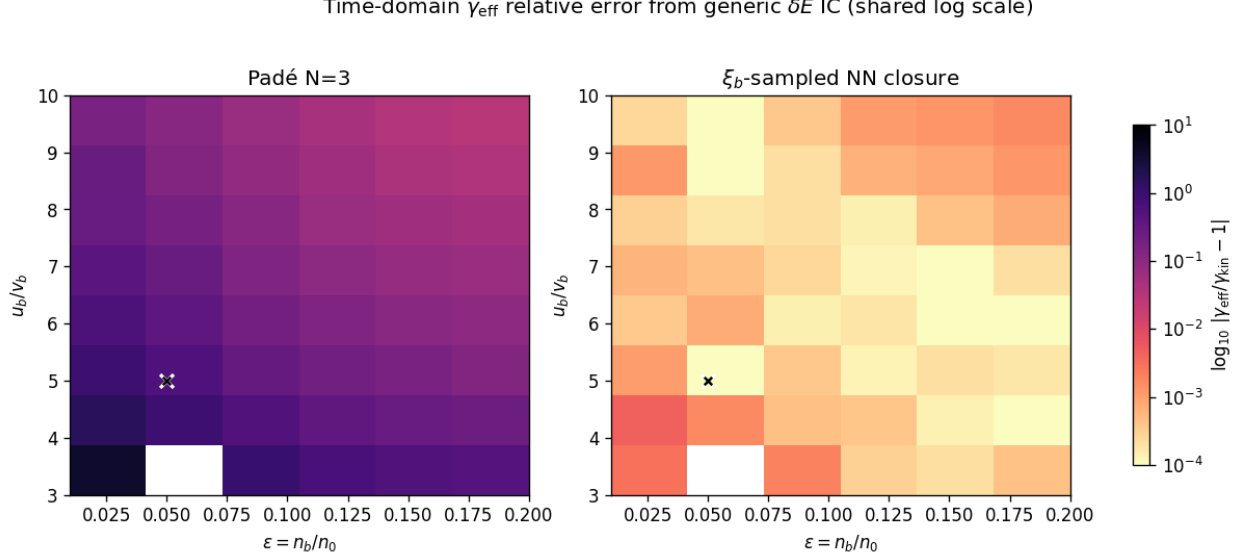


Figure 10: Time-domain γ_{eff} relative error from generic δE IC across the parameter grid: Padé (left) vs ξ_b -sampled NN closure (right) on the same shared log scale as Figure 6. The contrast against Figure 6 is the headline: same NN architecture, same fluid integrator, same generic IC, but now 10^{-4} – 10^{-3} across nearly the whole grid.

9 What did the NN actually learn?

Section 3 gives the structurally correct closure on a single eigenmode in closed form: $\alpha = r_1 \cdot r_2$. The NN was trained on ξ_b -sampled data, i.e. exclusively on on-manifold points. A natural question is whether the NN simply learned multiplication. The answer has two parts: yes on the manifold, no off it; and the off-manifold extrapolation is what makes the closure work.

On the eigenmode manifold. Sampling ξ_b uniformly in the training rectangle and evaluating $f_\theta(r_1(\xi_b), r_2(\xi_b))$ against $r_1 r_2$ gives a mean relative error of $\sim 3 \times 10^{-3}$, consistent with the NN’s training loss. So on the 2-real-dimensional submanifold the NN reproduces the bilinear product within training precision.

Off the manifold. Sampling r_1 and r_2 *independently* in the bounding box of the on-manifold cloud — points that do not correspond to any single ξ_b — gives a mean relative error of order one between $f_\theta(r_1, r_2)$ and $r_1 r_2$. The NN does not learn complex multiplication globally; it learned multiplication on the manifold and a smooth (but distinctly non-multiplicative) interpolation off it.

Why this matters: bilinear vs NN in time-domain. Figure 11 shows the consequence. At two operating points (canonical and marginal), we integrate four fluid systems – kinetic ground truth, NN closure, bilinear closure, Padé — from a generic δE IC. Top row: $|E(t)|$. The bilinear trajectory tracks Padé, not kinetic.

The bottom row isolates the closure function from any trajectory feedback: along the kinetic trajectory, we evaluate $\alpha_{\text{NN}}(r_1^{\text{kin}}, r_2^{\text{kin}})$ and $\alpha_{\text{bil}} = r_1^{\text{kin}} r_2^{\text{kin}}$ and compare both to the kinetic ground truth $U_3^{\text{kin}}/U_0^{\text{kin}}$. Two phases appear:

- *Early time* ($t \lesssim 20\omega_p^{-1}$ at canonical; $t \lesssim 50$ at marginal): the fluid state is a real superposition of eigenmodes, (r_1, r_2) sit off the manifold, and the bilinear product misses α_{kin} by 10^{-1} – 10^0 . The NN sits at $\sim 10^{-2}$ throughout.
- *Late time*: the unstable mode dominates, (r_1, r_2) relax onto the manifold, and the bilinear product becomes exact ($\sim 10^{-5}$ relative error). The NN plateaus at its $\sim 10^{-2}$ training residual.

Bilinear is the better closure function in the asymptotic single-mode limit; the NN is the better closure function in the early multimode phase. Because closure errors at early time seed spurious fluid eigenmodes that grow exponentially, early-time accuracy dominates the final γ_{eff} . The NN’s off-manifold smoothness is what locks the trajectory onto the kinetic eigenmode before spurious-mode contamination matters; hardcoded multiplication is the algebraic extension off the manifold and is the wrong extension.

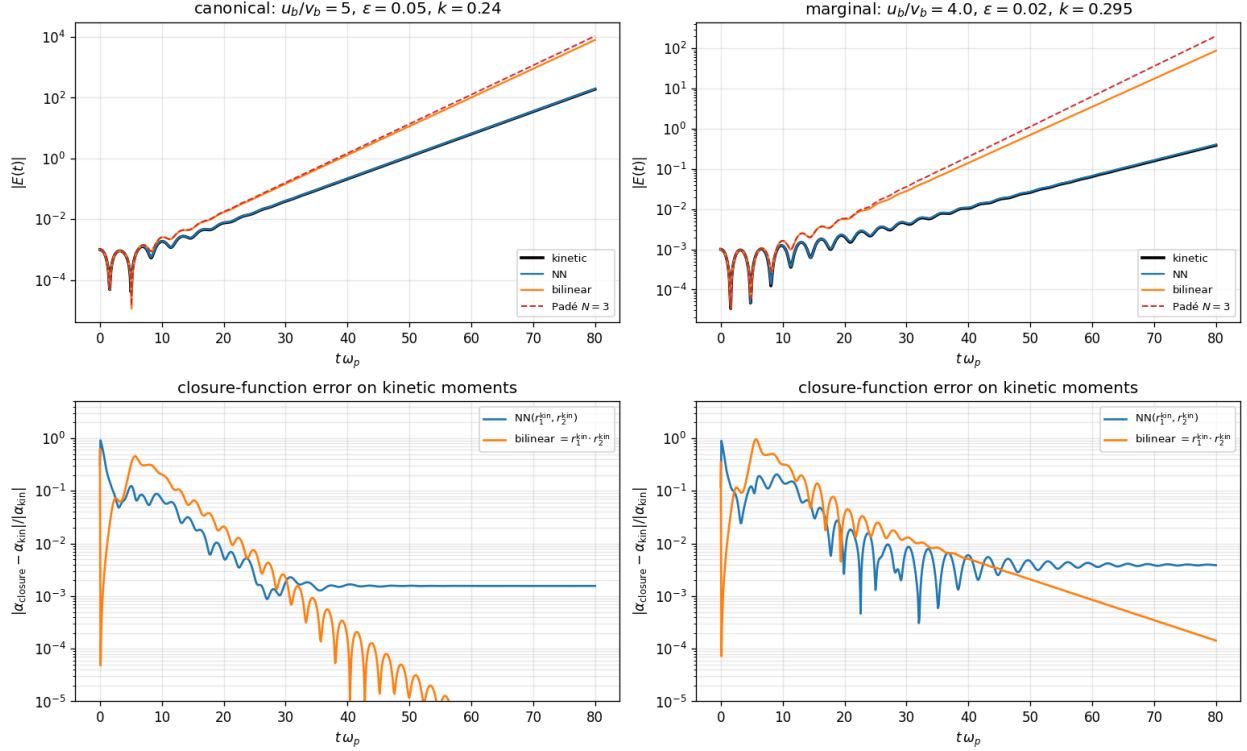


Figure 11: Off-manifold diagnostic at canonical (left) and marginal (right) operating points. *Top:* $|E(t)|$ for kinetic / NN / bilinear / Padé under a generic δE IC. Bilinear (orange) visibly tracks Padé, not kinetic. *Bottom:* closure-function error $|\alpha_{\text{closure}} - \alpha_{\text{kin}}|/|\alpha_{\text{kin}}|$ evaluated on the kinetic moments, isolating the closure function from trajectory feedback. NN: blue. Bilinear: orange. Early time ($t \lesssim 20$ canonical, $\lesssim 50$ marginal) the NN is one to two decades better than bilinear; once the unstable mode dominates, bilinear is exact and the NN plateaus at training residual.

Mechanism, in one line. On the eigenmode manifold the 3-moment Maxwellian closure is bilinear and exact. The NN reproduces that on the manifold and supplies a useful smooth extension off it — useful in the precise sense that it suppresses superposition cross-term error at early time and lets the trajectory relax onto the manifold. Linear closures (Padé/HP) miss both the on-manifold and the off-manifold structure; bilinear gets the on-manifold structure but fails on the off-manifold extension; the NN gets both.

10 Extension to the $N = 2$ even-moment closure

The analysis in Sections 8–9 focused on the $N = 3$ closure: predicting U_3 from (U_0, U_1, U_2) . We now examine the $N = 2$ level, where the closure target is U_2 from (U_0, U_1) .

System and exact per-mode formula. Truncating at two beam moments gives the four-dimensional state $y = (u, E, U_0, U_1)$ per Fourier mode k , with $\partial_t U_1 = -iku_b U_1 - ikU_2 + M_0 E$ and $M_0 = 1$. On a single eigenmode the $n = 0$ recurrence gives $U_1 = \xi_b U_0$ (zero-RHS since $M_{-1} = 0$), so $r_1 \equiv U_1/U_0 = \xi_b$ *exactly*. The $n = 1$ step has nonzero source $M_0 \hat{\Phi}$ and produces

$$\beta(\xi_b) \equiv \frac{U_2}{U_0} = \xi_b^2 - \frac{1}{Z'(\xi_b)}. \quad (11)$$

Because $r_1 = \xi_b$ the inference map is trivial, and the *direct- β* closure $U_2 = U_0 \beta(U_1/U_0)$ is simultaneously the exact on-mode formula and the natural off-manifold extension.

Structural contrast with $N = 3$. At $N = 3$ the $n = 2$ recurrence has zero RHS ($M_1 = 0$), giving $U_3 = r_1 r_2 U_0$: the bilinear identity $\alpha = r_1 r_2$ (Section 3). That identity is exact on eigenmode but produces $O(1)$ cross-term errors off it, motivating the NN. At $N = 2$ no bilinear identity exists ($M_0 \neq 0$), but the inference map is trivial and the exact β is already off-manifold-smooth: as the Langmuir mode grows to dominate, $r_1 \rightarrow \xi_b^{\text{kin}}$ and $\beta(r_1) \rightarrow \beta(\xi_b^{\text{kin}})$ exactly. Table 3 summarises the contrast.

Table 3: Structural comparison of $N = 2$ and $N = 3$ Maxwellian-beam closures.

	$N = 3$ (U_3 , odd)	$N = 2$ (U_2 , even)
Exact per-mode formula	$\alpha = \xi_b^3 - \xi_b/Z'(\xi_b)$	$\beta = \xi_b^2 - 1/Z'(\xi_b)$
Zero-parameter identity	$\alpha = r_1 r_2$ (bilinear)	none
NN input dimension	4 reals	2 reals
Inference ($r \rightarrow \xi_b$)	non-trivial	trivial: $r_1 = \xi_b$
Off-manifold	cross-terms $O(1)$; NN needed	smooth; exact formula sufficient

Padé $N = 2$. A linear closure $U_2 = a_0 U_0 + a_1 U_1$ Taylor-matched at $\xi_b \rightarrow \infty$ gives $\beta_{\text{Padé}}(\xi_b) = a_0 + a_1 \xi_b$ with $a_0 = -0.282$, $a_1 = 0.500i$ —a poor approximation to the nonlinear exact β (Figure 12).

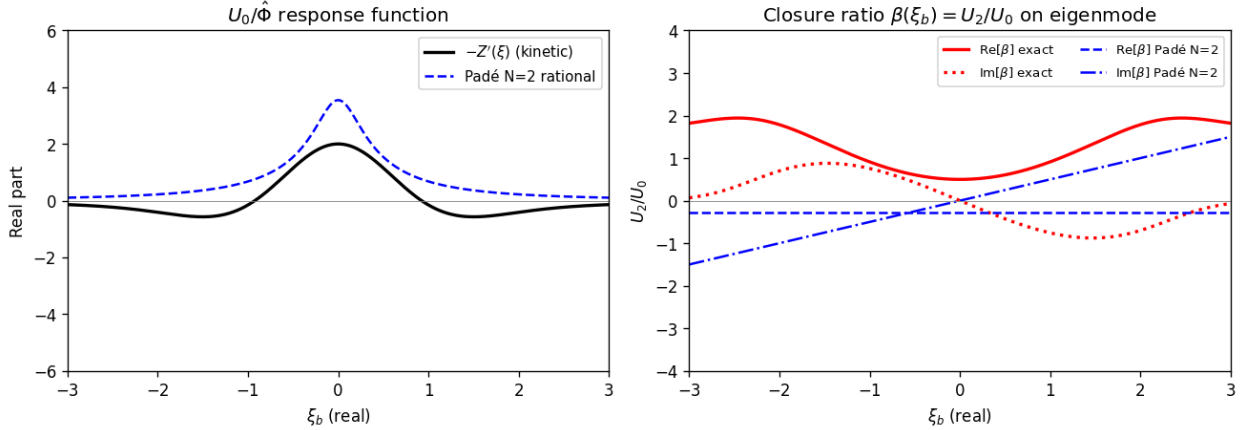


Figure 12: Left: kinetic response $-Z'(\xi_b)$ (solid) vs. Padé rational (dashed) along real ξ_b . Right: exact $\beta(\xi_b)$ (Re solid / Im dotted) vs. Padé-implied linear β (Re dashed / Im dash-dot). The linear β does not approximate the nonlinear exact β for $|\xi_b| > 0.5$.

Dispersion. Table 4 gives γ_{max} overshoot at four operating points. Direct- β recovers the kinetic rate to numerical precision everywhere; Padé $N = 2$ overshoots by +57% at the canonical point and +177% near the marginal corner.

Table 4: $\gamma_{\max}$ relative overshoot vs. kinetic, $N = 2$ closures.

u_b/v_b	ε	γ_{kin}	Padé $N = 2$	Direct β
5.0	0.05	+0.17028	+57%	0.00%
5.0	0.20	+0.28919	+28%	0.00%
8.0	0.05	+0.19991	+23%	0.00%
3.0	0.05	+0.11254	+177%	0.00%

Time-domain and off-manifold performance. Figure 13 shows that direct- β tracks the kinetic reference to within numerical precision under both a Langmuir-projected and a generic δE IC, while Padé $N = 2$ diverges immediately owing to its +55% growth-rate error. The off-manifold diagnostic (Figure 14) shows that Padé $N = 2$ plateaus at ~ 100 –200% closure error for all time; direct- β has a $\sim 10\%$ transient during the multi-mode phase ($t \lesssim 10$) and then decays to $< 10^{-5}$ as the Langmuir mode dominates and $r_1 \rightarrow \xi_b^{\text{kin}}$.

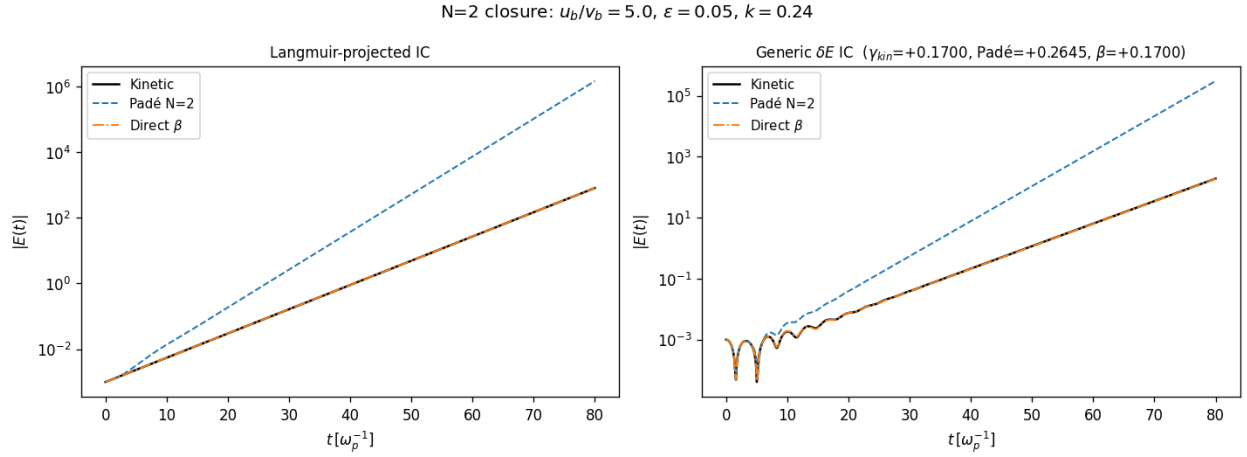


Figure 13: $|E(t)|$ at the canonical point $(u_b/v_b, \varepsilon, k) = (5, 0.05, 0.24)$ for kinetic (black), Padé $N = 2$ (blue dashed), and direct- β (orange) under a Langmuir-projected IC (left) and a generic δE IC (right). Fitted growth rates: kinetic +0.170, Padé +0.265, direct- β +0.170.

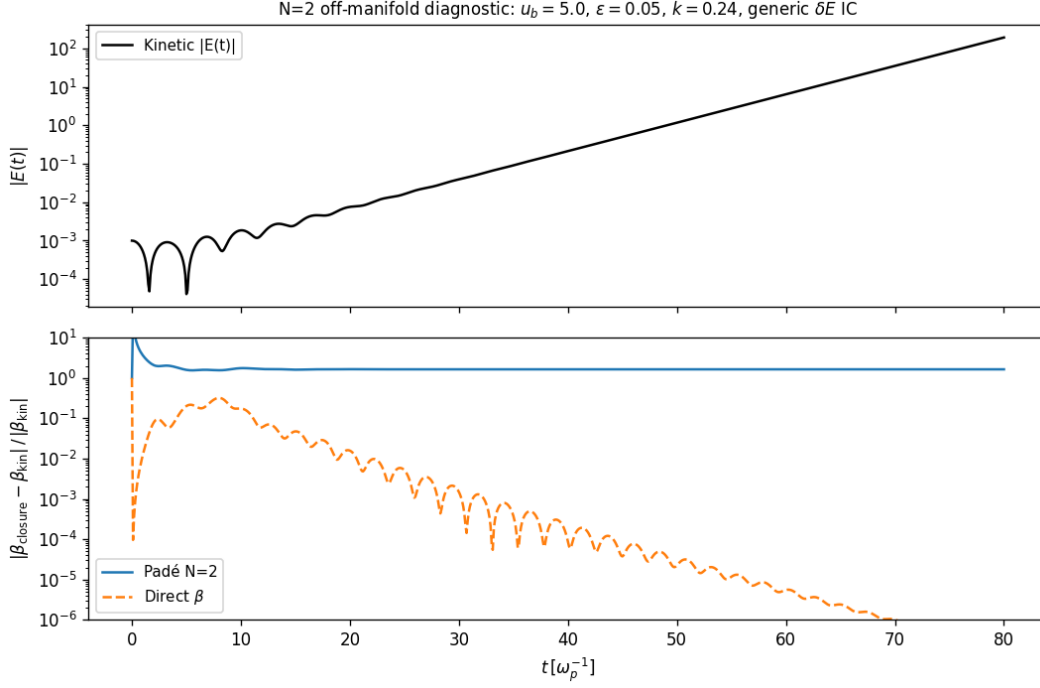


Figure 14: Top: $|E(t)|$ along the kinetic trajectory (generic δE IC). Bottom: closure error $|\beta_{\text{cl}} - \beta_{\text{kin}}|/|\beta_{\text{kin}}|$ evaluated at the kinetic moments. Padé $N = 2$ (blue solid) plateaus at $\sim 100\text{--}200\%$. Direct- β (orange dashed) peaks at $\sim 10\%$ during the multi-mode transient and decays to $< 10^{-5}$ once the Langmuir mode dominates.

Parameter sweep. Over the same $(u_b/v_b, \varepsilon)$ grid as Section 8 (Figure 15), direct- β stays within $\pm 0.5\%$ everywhere; Padé $N = 2$ overshoot ranges from $+9\%$ (hot dense beam) to $+800\%$ (cold sparse beam).

N=2 closures: γ_{eff} relative error vs kinetic (generic δE IC)

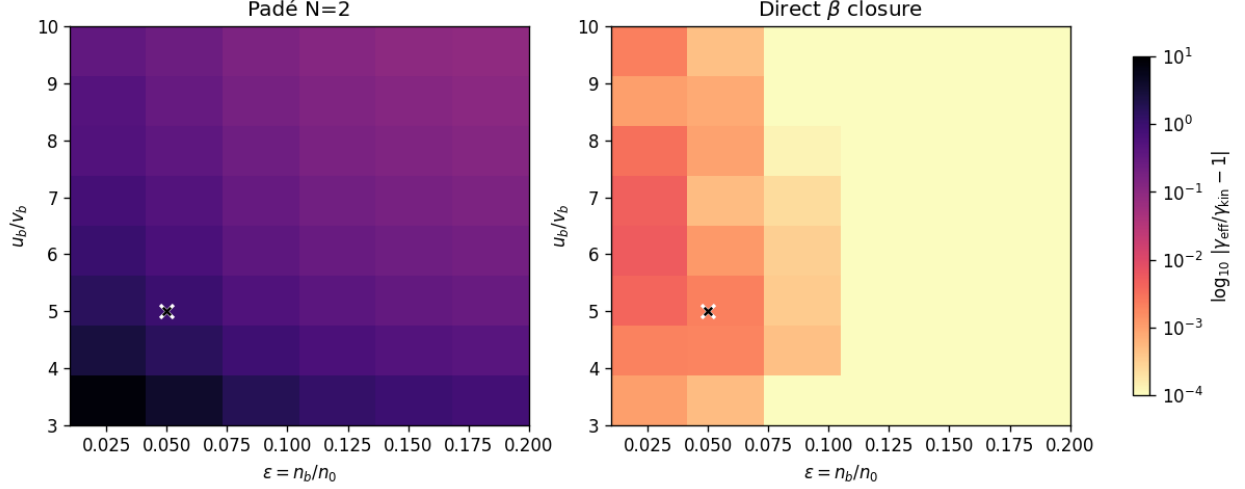


Figure 15: γ_{eff} relative error vs. kinetic on the $(u_b/v_b, \varepsilon)$ grid. Left: Padé $N = 2$, uniformly in 10^0 – 10^1 . Right: direct- β , $\lesssim 10^{-3}$ for $\varepsilon \geq 0.05$; the residual at the small- ε edge is fitting noise, not closure error.

11 Generalization: Landau damping on a single warm Maxwellian

The trained closure $f_\theta(r_1, r_2) \rightarrow \alpha$ depends only on the Maxwellian linear response: nothing in the training set references u_b , ε , or k . So the same checkpoint should work for any 3-moment fluid system on a Maxwellian equilibrium, not just BoT. This section confirms it by computing Landau damping of Langmuir waves on a single warm Maxwellian – a textbook benchmark unrelated to the BoT instability.

Setup. Strip the BoT system to a single warm species (drop the cold bulk equations, set $\varepsilon = 1$, $u_b = 0$):

$$\begin{aligned}\partial_t E &= -U_1, \\ \partial_t U_0 &= -ikU_1, \\ \partial_t U_1 &= -ikU_2 + E \quad (M_0 = 1), \\ \partial_t U_2 &= -ikU_3 \quad (M_1 = 0, \text{ closure}).\end{aligned}$$

The kinetic Langmuir dispersion in this normalization is $k^2 = Z'(\omega/k)$, related to the standard $k\lambda_D$ convention by $k_{\text{code}} = \sqrt{2}k\lambda_D$ (BoT-paper variance is $1/2$, so $v_{th} = 1/\sqrt{2}$).

Method. For each $k\lambda_D$:

1. Solve $k^2 = Z'(\omega/k)$ for the kinetic Langmuir mode (Newton from the Bohm-Gross seed) – analytic ground truth.
2. Project initial state onto the kinetic Langmuir eigenvector: $(E, U_0, U_1, U_2) = (U_1/i\omega, 1, \xi, \xi^2 - 1/Z'(\xi))$ with $\xi = \omega_{\text{kin}}/k$.

3. Integrate the NN-closed and Padé-closed fluid systems in time, fit (ω_r, γ) from late-time log-magnitude slope and real-part zero crossings.

Projecting onto the kinetic eigenvector isolates the Langmuir mode – without it, the fluid system’s spurious closure eigenmodes (which exist because there is no cold-bulk grounding) dominate the late-time signal.

Result. Figure 16 shows $(\omega_r, |\gamma|)(k\lambda_D)$ on $k\lambda_D \in [0.30, 0.60]$.

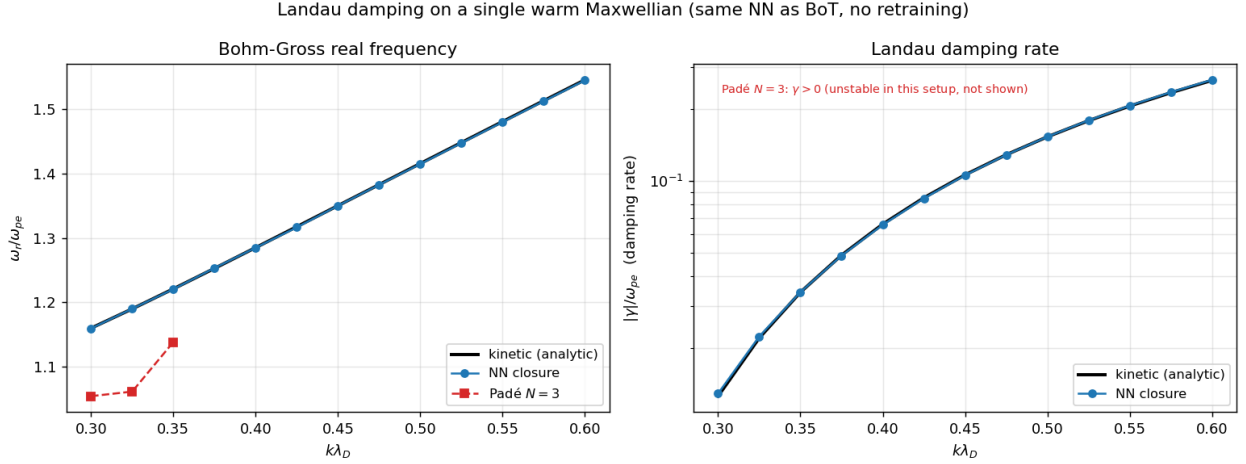


Figure 16: Landau damping of Langmuir waves on a single warm Maxwellian, using the same NN checkpoint trained for BoT. *Left*: Bohm-Gross real frequency. *Right*: damping rate, log scale. The NN closure reproduces the kinetic ground truth (textbook values: $\gamma/\omega_{pe} = -0.0126$ at $k\lambda_D = 0.3$, -0.1534 at $k\lambda_D = 0.5$, etc.) to better than 1% across the range. Padé $N = 3$ in this setup gives the wrong frequency and is unstable ($\gamma > 0$); not shown on the log-damping axis. See text.

The NN curve overlays the kinetic curve in both panels. Representative values:

$k\lambda_D$	kinetic γ/ω_{pe}	NN γ/ω_{pe}	rel. error
0.30	-0.0126	-0.0129	2.4%
0.40	-0.0661	-0.0657	0.6%
0.50	-0.1534	-0.1535	0.07%
0.60	-0.2641	-0.2655	0.5%

Why Padé fails here. The same Hunana-style Padé closure that gave $\sim 30\%$ overshoot on BoT γ_{eff} produces an *unstable* fluid system on the pure warm Maxwellian: $\gamma > 0$ even when initialized on the kinetic Langmuir eigenvector. This is a known consequence of Taylor-matching closures in systems without cold-fluid grounding; Hammett-Perkins-style frequency-axis filtering can fix it but is a separate construction. The NN closure has no such problem because the per-mode bilinear identity it learned (Section 3) gives the exact Langmuir eigenmode by construction: the closed fluid system’s Langmuir eigenvalue equation reduces to $Z'(\omega/k) = k^2$ identically.

What this means. The closure object stored in `naive_mlp.eqx` is a learned Maxwellian linear-response function. It is reusable for any 3-moment fluid integrator with a Maxwellian equilibrium and ξ_b in the trained rectangle. BoT is one instance, Landau damping on a warm Maxwellian is

another, and the same logic extends to ion-acoustic waves with adiabatic electrons, drift waves with adiabatic species, or any model where the closure is "evaluate Z' of an effective ξ_b ". A separate trained model is needed only when the equilibrium is non-Maxwellian (kappa, flat-top, trapped plateau) or when going beyond 3 moments (the per-mode identity at higher order is not just a product because $M_2 \neq 0$).

Landau-damped BoT: Padé structural failure. We return to the full BoT configuration but move into the regime where the minority beam provides Landau *damping* rather than instability. This requires $u_b < \omega_r/k$, so that $\text{Im}(\xi_b) < 0$ for the relevant mode. Three test points are $(u_b, k) = (2.0, 0.30)$, $(1.5, 0.40)$, $(1.0, 0.50)$ with $\varepsilon = 0.05$. The resulting ξ_b values $(1.93 - 0.14i$ to $1.25 - 0.34i)$ lie well inside the training rectangle.

Figure 17 shows the growth rates from dispersion for all $N = 2$ closures. Padé $N = 2$ predicts spurious instability ($\gamma > 0$) at every test point. The reason is structural: the Taylor-series approximant is anchored on the real ξ_b -axis and cannot represent the Landau-pole contribution that lives in the lower half-plane ($\text{Im}(\xi_b) < 0$). The direct- β and NN(r_1) closures both correctly predict negative γ because they evaluate the Faddeeva function at the actual complex ξ_b . This failure is not a matter of closure accuracy—it is a fundamental limitation of any polynomial or rational approximation matched by Taylor expansion on the real axis.

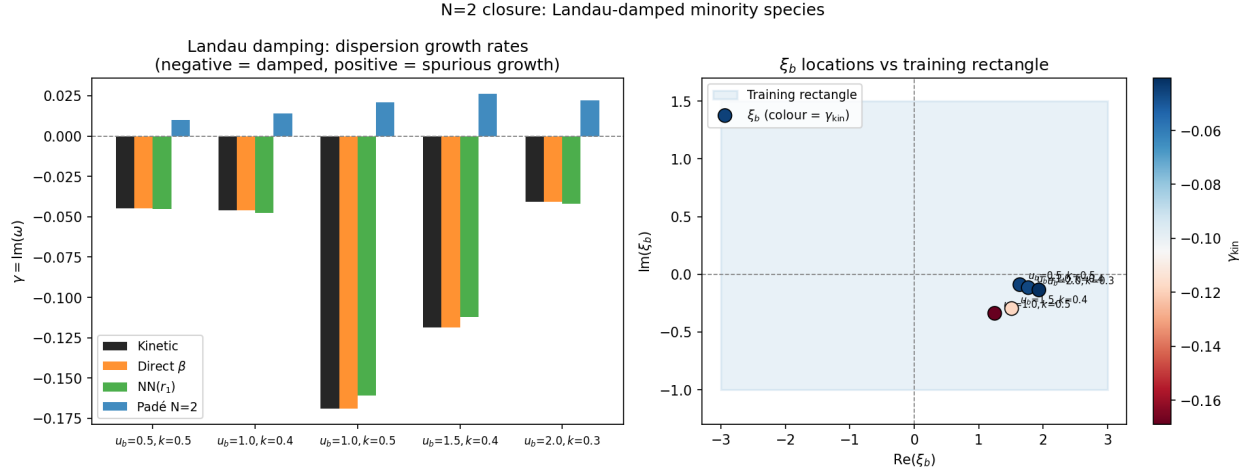


Figure 17: Analytic growth rates $\gamma = \text{Im}(\omega)$ at three Landau-damped BoT test points (negative = damped). Padé $N = 2$ (blue) predicts positive γ at every point, a qualitative failure. Direct- β (orange) and NN(r_1) (green) correctly predict negative γ . Right panel: ξ_b locations in the complex plane; all points lie within the training rectangle.

Comparing $N = 2$ and $N = 3$ closures off eigenmode. To compare closures across both truncation levels under multi-mode conditions, we integrate from a generic δE IC at each Landau-damped test point and compare to the kinetic ground truth. To first establish a clean baseline, Figure 18 shows $|E(t)|$ under an eigenmode-projected IC (top row) and a generic δE IC (bottom row), for all available $N = 2$ and $N = 3$ closures.

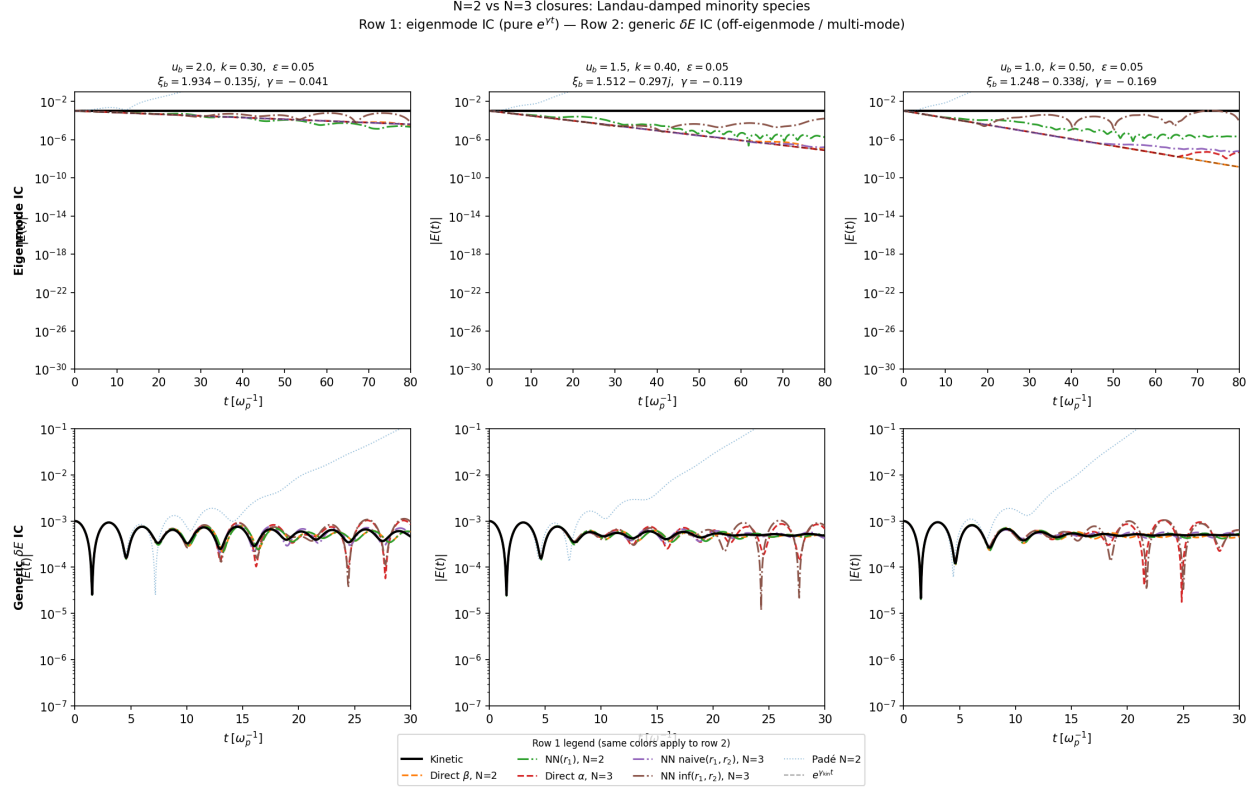


Figure 18: Top row: eigenmode-projected IC. Direct- β N=2 and direct- α N=3 (exact on eigenmode) decay cleanly along the reference slope $e^{\gamma_{\text{kin}} t}$. Both NNs also decay but level off at later times as approximation error accumulates. The kinetic $N_v=96$ curve is nearly flat: $O(10^{-14})$ mixing with van Kampen quasi-continuum modes dominates after ~ 10 e-folds. Bottom row: generic δE IC; kinetic (black) is ground truth.

Table 5 gives the mean relative error $\bar{\epsilon} = \langle |E_{\text{cl}} - E_{\text{kin}}| / |E_{\text{kin}}| \rangle$ over $t \in [5, 30]$ for the generic-IC runs. Four patterns stand out. *NN naive(r_1, r_2) N=3 is the best or joint-best*: the second input $r_2 = \langle \beta(\xi) \rangle$ carries genuine off-eigenmode information about the spread of ξ across modes. *NN inference N=3 is the worst* (worse than even the parameter-free direct- α): the inference bottleneck forces a single $\hat{\xi}_b$ inconsistent with multi-mode r_1 and r_2 , and the exact Faddeeva at the compromised $\hat{\xi}_b$ amplifies the error. *Direct- α N=3 is the second worst*: α 's larger curvature amplifies the Jensen error relative to β . *Direct- β N=2 and NN(r_1) N=2 are comparable*, as both use only r_1 .

Table 5: Mean relative error vs. kinetic under generic δE IC, $t \in [5, 30]$. “Naive” NN maps $(r_1, r_2) \rightarrow \alpha$ directly; “Inference” NN maps $(r_1, r_2) \rightarrow \hat{\xi}_b \rightarrow \alpha(\hat{\xi}_b)$.

Case	Dir. β N=2	NN(r_1) N=2	Dir. α N=3	NN naive N=3	NN inf. N=3
$u_b = 2.0, k = 0.30$	0.24	0.24	0.37	0.13	0.42
$u_b = 1.5, k = 0.40$	0.13	0.14	0.33	0.10	0.45
$u_b = 1.0, k = 0.50$	0.12	0.10	0.44	0.11	0.50

12 Discussion

Where the win comes from. Both wins — accuracy and cost — trace to the same observation: the closure problem is parameterised by a single complex variable ξ_b . Simulation training generates (ratios, α) pairs via the parameterisation $(u_b/v_b, \varepsilon, k, \text{IC}, t) \mapsto \xi_b$, which is many-to-few and covers only a 1-parameter family of ξ_b values (the unstable eigenmode), at the cost of a kinetic integration per operating point. Frequency-domain training generates the same pairs via the identity $\xi_b \mapsto (\text{ratios}, \alpha)$, which is one-to-one, covers whatever subset of $\mathbb{C}$ we choose, and reduces dataset generation to a Faddeeva evaluation. The 2D rectangle is the natural, dense, problem-agnostic training distribution because moment ratios from the 5-d fluid system’s other (damped) eigenmodes also land in this rectangle.

The sim-trained closure is not bad — $\sim 10\%$ on γ_{eff} across most of the grid is a real improvement over Padé. The honest framing is that the closed-form approach is strictly better on every axis that matters: tighter error (two decades), cheaper data (two orders of magnitude), problem-agnostic (no operating-point sweep), and trivially re-trained when the operating regime shifts.

The same logic applies to other moment closures whose linear-response function is one-parameter (drift-kinetic with one resonance, weakly collisional fluid models on a Maxwellian background): if there is an analytical $\xi_b \mapsto \alpha$ map, frequency-domain sampling is strictly better than simulation sampling at fixed dataset size.

What is not solved. The training is entirely linear-physics: $\alpha(\xi_b)$ is derived assuming the beam perturbation is linear about the Maxwellian f_{b0} . Capturing trapping, plateau formation, and saturated BoT requires kinetic training data and is outside this scope. The ξ_b -rectangle covers the unstable BoT regime by construction; the marginal corner $(u_b/v_b \lesssim 3, \varepsilon \lesssim 0.02)$ pushes ξ_b to the rectangle’s lower edge and shows mild degradation.

Relation to the Padé / Hunana hierarchy. Hunana et al. (2019) construct rational closures of $Z'(\zeta)$ up to 8 poles and prove asymptotic convergence with increasing order. The NN closure trades the rational-order ceiling for NN expressivity; the present work shows the fluid response problem itself isn’t the obstacle — training-data design is.

Inference vs. naive NN architecture. At $N = 3$, two distinct NN parameterisations have been tested. The *inference* architecture maps $(r_1, r_2) \rightarrow \hat{\xi}_b$ and then evaluates $\alpha(\hat{\xi}_b)$ via the exact Faddeeva function. The *naive* architecture maps $(r_1, r_2) \rightarrow \alpha$ directly.

On the single-eigenmode manifold, both are equivalent by construction: (r_1, r_2) determines ξ_b uniquely, so the inversion is exact and the two designs converge to the same prediction. Off eigenmode—as quantified in Table 5—they diverge sharply. The inference NN is the *worst* performer (42–50% mean relative error), worse than the parameter-free direct- α . The reason is structural: in a multi-mode state, $r_1 = \langle \xi \rangle$ and $r_2 = \langle \beta(\xi) \rangle$ cannot simultaneously be consistent with any single $\hat{\xi}_b$, so the NN settles on a compromised value, and the exact Faddeeva applied there amplifies rather than corrects the error. The inference design embeds the wrong physical assumption—that the beam is always in a single eigenmode—directly into the architecture.

The naive NN carries no such assumption. Its direct $(r_1, r_2) \rightarrow \alpha$ map learns a smooth, non-multiplicative function that implicitly approximates $\langle \alpha(\xi) \rangle$ from partial moment information, without committing to any single-resonance picture. This makes it the most robust closure when multiple modes are simultaneously present, which is the generic situation in a fluid simulation.

The practical recommendation is therefore: *train the NN as a direct map from moment ratios to the closure coefficient, not as an inference engine for an intermediate physical parameter.* The

exact Faddeeva is beneficial only when ξ_b is known exactly; when it must itself be inferred from multi-mode moment data, the inversion error propagates nonlinearly.

Even-moment closures. As shown in Section 10, the $N = 2$ closure illustrates the complementary case where no NN is needed at all: the inference map is trivial ($r_1 = \xi_b$), the exact β is analytically available, and its off-manifold behavior is already superior to any linear approximation. The interesting NN problem is thus restricted to odd-moment levels (or higher even levels where the general per-mode formula involves multiple Maxwellian moments and the off-manifold behavior may again degrade).

Code

The accompanying `bot/` package implements all figures and runs:

- `closures/pade.py` — Padé closure coefficients and the closed fluid response.
- `closures/naive_nn.py` — NN closure (MLP $\rightarrow \alpha$), closed-form ξ_b -sampling dataset, training.
- `closures/u2.py` — $\beta(\xi_b)$ exact formula, MLP_{*u2*} (2-input NN for β), training utilities.
- `train_simdata.py` — sim-data NN training (single- and broad-sweep).
- `kinetic.py`, `fluid.py` — kinetic ground truth and two-fluid eigenvalue system.
- `closed_loop.py` — time-domain integration for all closures; eigenmode IC helpers for $N = 2$, $N = 3$, and kinetic.
- `sweep.py`, `sweep_timedomain.py` — 2D parameter scans.
- `figures.py` — $N = 3$ plots.
- `figures_u2.py` — $N = 2$, Landau damping, and $N = 2/N = 3$ comparison plots.